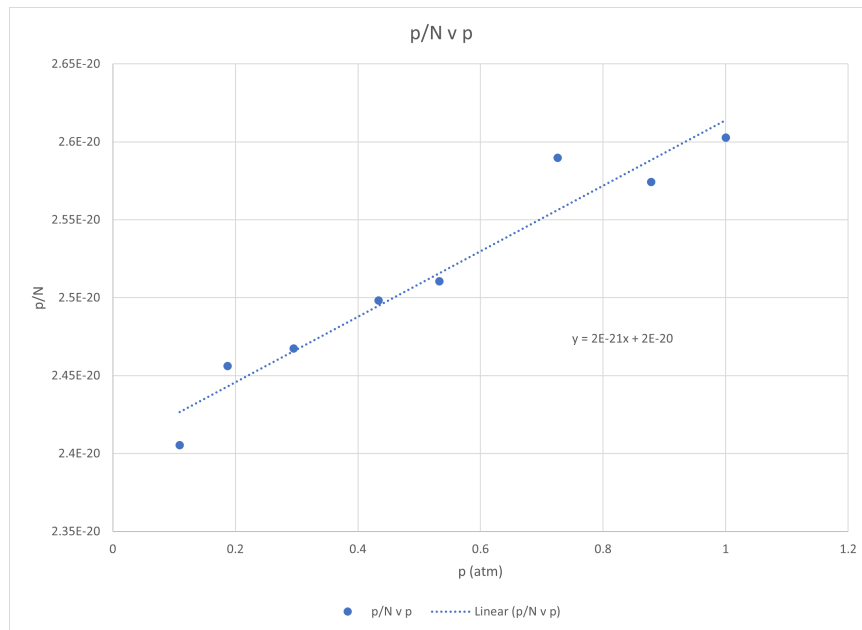


6. Problem 6 (7.17)

a.

p (atm)	N _{O₂} (abs)	p/N
0.1092	4.54	$2.40529 \cdot 10^{-20}$
0.1874	7.63	$2.45609 \cdot 10^{-20}$
0.2951	11.96	$2.46739 \cdot 10^{-20}$
0.4337	17.36	$2.49827 \cdot 10^{-20}$
0.533	21.23	$2.5106 \cdot 10^{-20}$
0.7259	28.03	$2.58973 \cdot 10^{-20}$
0.8783	34.12	$2.57415 \cdot 10^{-20}$
1	38.42	$2.60281 \cdot 10^{-20}$



$$\text{slope} = \frac{1}{N_{sat}}$$

$$N_{sat} = \frac{1}{2.10 \cdot 10^{-21}}$$

$$N_{sat} = 4.75 \cdot 10^{20} \text{ molecule/g}$$

b.

$$4.75 \cdot 10^{20} \text{ molecule/g} \cdot 19.8 \cdot 10^{-20} \text{ m}^2 = 94.1 \text{ m}^2/\text{g}$$

c.

$$N_{sat} = \frac{94.1 \text{ m}^2/\text{g}}{32 \cdot 10^{-20} \text{ m}^2} = 2.94 \cdot 10^{20} \text{ molecule/g}$$

7. Problem 7 (7.23)

$$K_{eq,1} = \frac{[\text{NO}] [\text{NO}_2] [\text{H}_2\text{O}]}{[\text{HNO}_2]^2}$$

$$K_{eq,2} = \frac{[\text{N}_2\text{O}_4]}{[\text{NO}_2]^2}$$

$$\frac{d[\text{NO}_3^-]}{dt} = k_3 [\text{N}_2\text{O}_4] [\text{H}_2\text{O}]$$

$$[\text{N}_2\text{O}_4] = K_{eq,2} [\text{NO}_2]^2$$

$$\frac{d[\text{NO}_3^-]}{dt} = k_3 K_{eq,2} [\text{NO}_2]^2 [\text{H}_2\text{O}]$$

$$[\text{NO}_2]^2 = \left(K_{eq,1} \frac{[\text{HNO}_2]^2}{[\text{NO}] [\text{H}_2\text{O}]} \right)^2$$

$$\frac{d[\text{NO}_3^-]}{dt} = k_3 K_{eq,2} K_{eq,1}^2 \frac{[\text{HNO}_2]^4}{[\text{NO}]^2 [\text{H}_2\text{O}]}$$

$$\frac{d[\text{NO}_3^-]}{dt} = -\frac{1}{3} \cdot \frac{d[\text{HNO}_2]}{dt}$$

$$-3 \frac{d[\text{NO}_3^-]}{dt} = \frac{d[\text{HNO}_2]}{dt}$$

$$\boxed{\frac{d[\text{HNO}_2]}{dt} = -3k_3 K_{eq,2} K_{eq,1}^2 \frac{[\text{HNO}_2]^4}{[\text{NO}]^2 [\text{H}_2\text{O}]}}$$